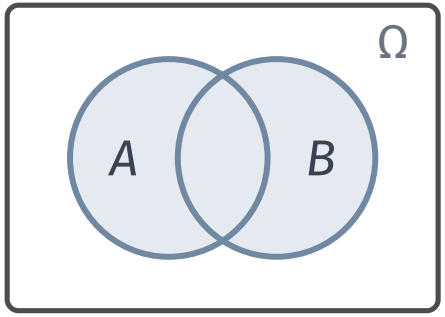
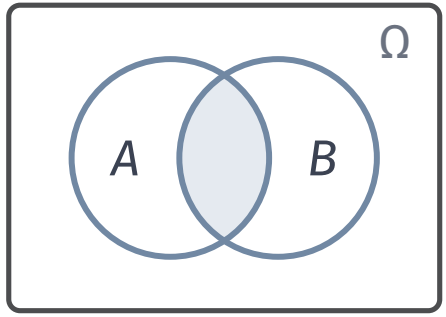
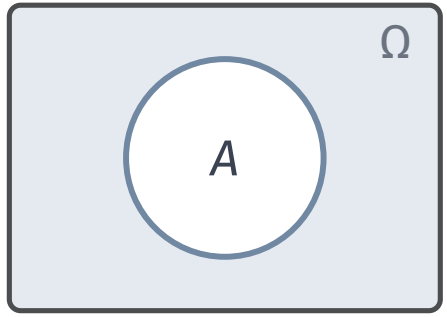


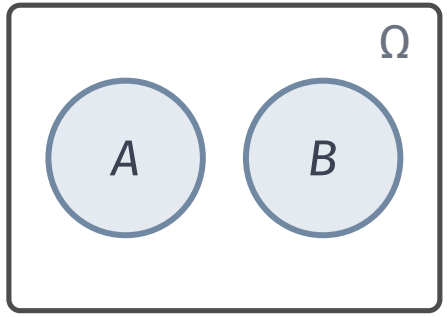
(a) $A \cup B$



(b) $A \cap B$



(c) A^c



(d) $A \cap B = \emptyset$